Assignment 6

Q1. Simulate the producer-consumer problem using shared memory and process synchronization. Implement the following classical algorithms:

1. Dekker's Algorithm
2. Lamport's Algorithm
3. Peterson's Algorithm

Q2. Given a finite buffer, you need to implement solution for Producer Consumer Problem for the following versions

(1) Single Producer – Single Consumer

(2) Single Producer – Multiple Consumers

(3) Multiple Producers – Single Consumer

(4) Multiple Producers – Multiple Consumers

through use of Peterson's algorithm

Q3. Create a program that simulates a simple bank with multiple accounts and multiple clients making deposits (thread1) and withdrawals (thread2) concurrently. The goal is to ensure that account balances remain consistent even with concurrent operations. Use mutex locks to implement this.

Q4. Write a program in C to create a child process using fork() system call, parent and child shares a variable int VAR = 10; parent and child can execute concurrently, parent increments the value of VAR by 2 and child decrements the value by 2. Synchronize both the process by using semaphore such that child should always execute before parent.

Q5. Write a C programs to implement multithreading where first thread calculates the sum of the elements of shared data (int data [10]), another thread finds the maximum value, and the third thread finds the minimum value. The main thread waits for these threads to finish and prints their results.